CELNAV quick reference

Menu: run `celnav`. **Everything also works from the command line.**

`celnav dr "35 10.4 N" "040 20.1 W" 245 11`	set DR, course true, speed
`celnav plan`	twilight, what is up, best three
`celnav sight "2026-08-29 07:30:00" Dubhe C "19 32.1" 1.5 3.0`	UT, body, limb, Hs, index error, eye height
`celnav fix`	reduce the log and plot the fix
`celnav noon "2026-10-21" "28 51.4"`	latitude from the sun at its highest
`celnav polaris "UT" "50 18.5"`	latitude from Polaris, all night long
`celnav compass "UT" sun 112.5 6.0`	compass error and deviation
`celnav devcard "DATE" sun -2 "000,10:00:00,147.0; ..."`	swing → deviation card
`celnav current "50 00 N" "005 00 W" "50 03 N" "004 56 W" 4`	set and drift
`celnav ep "50 00 N" "005 00 W" 090 6 4 180 1`	DR run, and the EP
`celnav table 50 -11 329`	a Pub. 249 page
`celnav log` / `celnav clear`	list / empty the sight log
`celnav doctor`	environment check and self test
`celnav learn`	lessons, walkthrough, drills, sandbox
`celnav drill red`	one marked drill (corr, alm, red, int, full, fix)
`celnav night` / `day` / `plain`	colour mode

Menu keys: 1 sight · 2 fix · 3 plan · 4 almanac · 5 compass · 6 DR · 7 log · 8 settings · 9 learn · s stars · t test · d check · c colour · h help · q quit

Order of work

1. **DR** — position, course made good, speed. Without course and speed the sights are not advanced.
2. **Plan** — before twilight. Note the three bodies and their approximate altitudes so you can find them fast.
3. **Shoot** — record UT to the second. *Four seconds is one mile of longitude.*
4. **Enter** each sight. Limb: `L` lower, `U` upper, `C` centre (stars and planets).
5. **Fix** — read the residuals and the geometry line before you use it.
6. **Clear** the log.

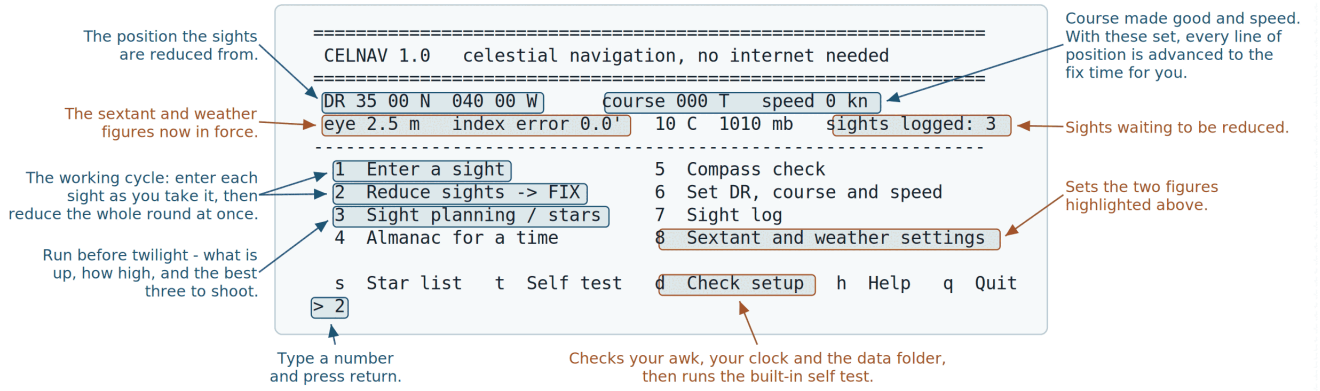
Reading the answer

- **Intercept** — miles from your DR toward (+) or away (−) from the body.
- **Residual** — how far each line of position misses the fix. This is your observing error. Under a mile is good work.
- **RMS scatter** — the same thing for the round as a whole.
- **GEOMETRY** — how far one arcminute of sight error moves the fix. Under 1.6 nm is a sound cut; over 3 nm is a weak one and CELNAV says so.

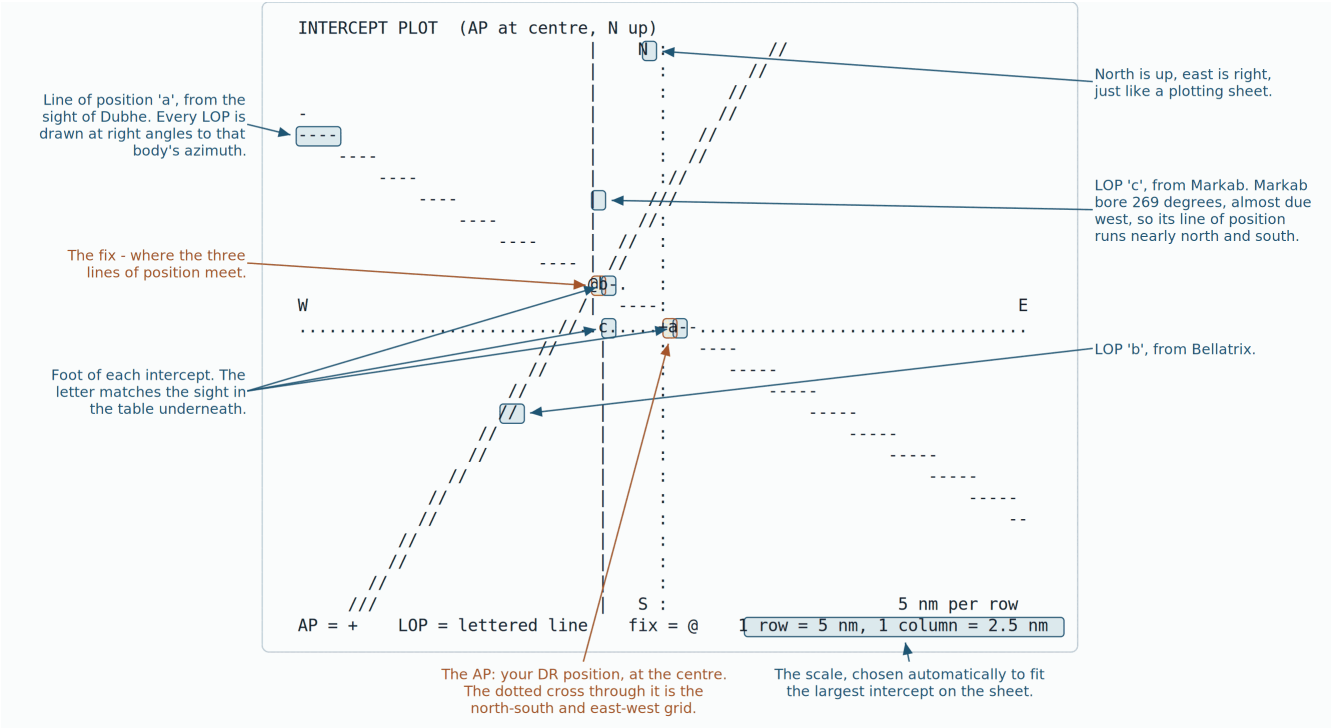
Formats

```
Angles     35 10.4 N     040 20.1 W     -35.1733     040:20.1W
Altitude   32 14.6       (degrees and decimal minutes)
Time       2026-08-29 07:30:00 UT   (or 073000 in the menu, today's date assumed)
```

The screen



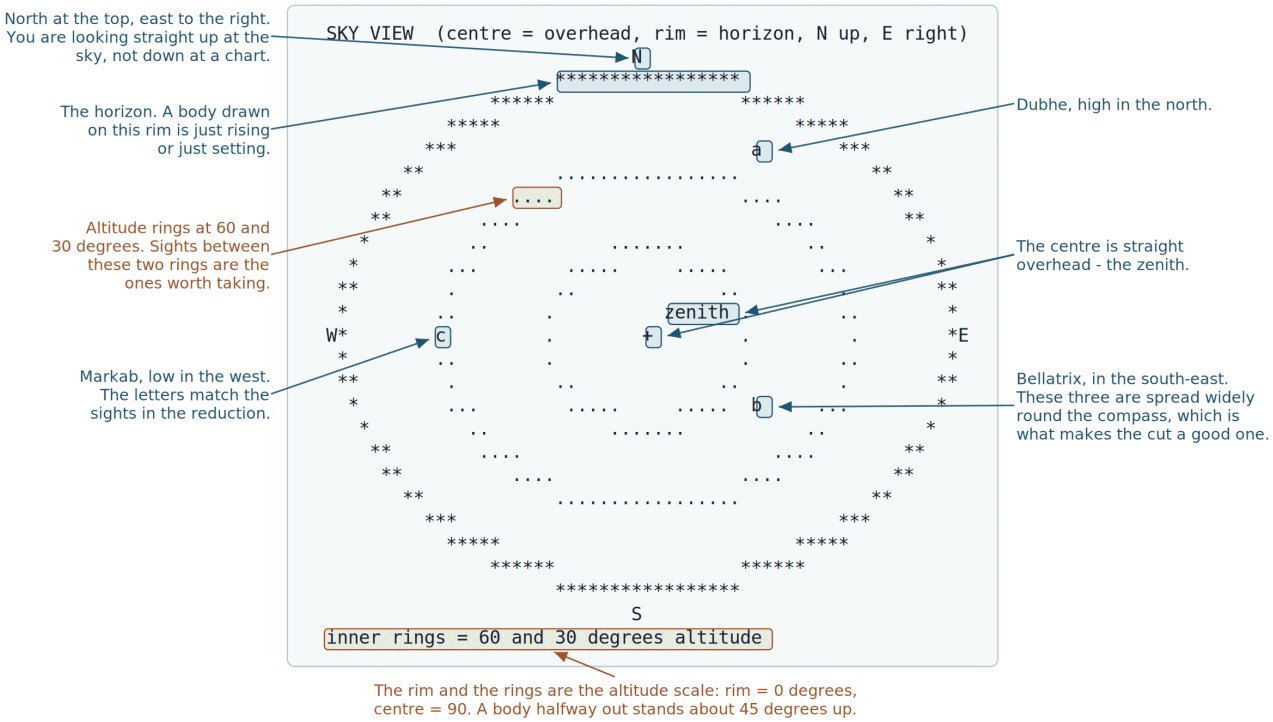
The menu. The two status lines always show what the next reduction will be based on.



The intercept plot. Your DR at the centre, one lettered line for each sight, the fix where they cross.

Reading the plot: `+` is your DR at the centre of the sheet · each lettered line is a line of position, drawn at right angles to that body's azimuth · the letter sits at the foot of its intercept · `@` is the fix · the scale is printed at the bottom right and is chosen to fit the largest intercept.

The sky view



The sky view from `celnav plan` — centre is straight overhead, the rim is the horizon, north is up. Bodies spread widely round the compass give the tightest fix.

When something is wrong

Symptom	First thing to check
Fix far from DR with large residuals	the **time** of each sight — a wrong digit in the seconds moves longitude only
“Single line of position only”	two sights on nearly the same azimuth — add a body 40–120° away
“no awk with trigonometric functions”	`apk add gawk` on iSH, `pkg install gawk` on Termux
Self test fails	the file transfer truncated — check `cksum celnav`

Colour

`c` in the menu, or `celnav day` / `night` / `plain`.

- **day** — white on black, highlights in green.
- **night** — deep red on black, highlights in bright red. No green at night: twenty minutes of dark adaptation is not worth a prettier screen.
- **plain** — no escape sequences; chosen automatically when output is not a terminal.

Learning it

`celnav learn`, or menu item 9.

- **Lessons** — thirty-three, in nine modules, from dead reckoning and set and drift (a preliminary module you can test out of) through what a sight measures to choosing the bodies, checking the compass against them, taking latitude directly from the sun at noon and Polaris at any hour, handling the moon and the planets, and working a sight by Pub. 249 with the exact answer beside it. Each ends with a question; answer it and the lesson is ticked off.
- **Walkthrough** — one real sight in ten steps, every line explained as it is computed.
- **Drills** — problems built from the real almanac and marked, with the full working shown afterwards. Types: `corr`, `alm`, `red`, `int`, `full`, `fix`.
- **Sandbox** — change latitude, declination or LHA and watch the navigational triangle move.

Rules of thumb

- Prefer bodies between **15° and 70°** of altitude.
- Three bodies about **120° apart** in azimuth give the tightest fix.
- The **star horizon** is usable between civil and nautical twilight — a window of roughly twenty minutes.
- A sight of the Sun at any altitude gives a line; two of them, an hour or more apart, give a running fix.

The camera sight (browser build)

An altitude from a photograph, the device's camera standing in for the sextant. Manual section 13 has the whole of it; this is what to remember on deck.

Step	
1 Photograph	sun or moon **with the horizon in the same frame**
2 Horizon	tap the horizon, then *Find the edge*
3 Body	tap the body, then *Find the edge*
4 Sight	body, UTC time, assumed latitude and longitude
5 Plot	keep the sight; a second body gives a fix

- **Do not zoom.** A digitally zoomed frame is a crop upscaled, the plate scale may be wrong, and a wrong plate scale gives a confident sight in the wrong place. Shoot at the lens's own focal length and crop afterwards.
- **Calibrate first, from a star field** — three stars at least, four better, at **different distances from the frame's centre**, shot **high up**. No time and no position are needed; refraction near the horizon is what spoils it.
- **Uncalibrated is an exercise, not a position.** Barrel distortion at `k1 = -0.05` costs up to **44 arcminutes — forty-four miles.**

- **A residual over 3 arcminutes is refused.** Usually a misidentified star, or a cropped or zoomed frame.
- **Tap roughly, then let it find the edge.** One pixel is **1.18 arcminutes, about 1.2 miles**; a fingertip covers about forty of them. If a refinement jumps a long way it has found a wake or a cloud edge, not the horizon.
- **Second body at least 40° away in azimuth.** Parallel lines are not a fix — one line still tells you that you are somewhere along it.